

Can we recreate the sound of a church in real time?

This thesis investigates whether it is possible to recreate spatial audio inside a large church, giving the illusion of sound bouncing off the walls and into your ears. If successful, anyone with a pair of headphones and access to a modern phone or computer could experience the acoustics of the church, an auditory experience that would otherwise be beyond their reach. However, doing this convincingly and in real time turns out to be more challenging than one might think.

By recording audio inside a church, we can capture what makes that room sound unique, both the direct sound that reaches your ears immediately and the echoes that arrive later after reflecting off walls, ceilings and other surfaces. The main challenge is processing all of this quickly enough that it feels natural. In theory, this could make it possible to filter your voice during a phone call so that someone wearing headphones hears it as if you were both standing inside a church.

Three different methods were proposed and tested. The first was simple and straightforward to implement, but too slow to meet the demands of real-time audio. The second achieved the same result much faster, and proved quick enough for real-time processing. The third compressed the audio data very efficiently and greatly reduced storage requirements, but still fell short of real-time performance. This made the second method the clear winner, and the only one able to achieve the main goal of the project.

One challenge the study encountered was how difficult it is to measure how convincing an audio effect sounds without involving human listeners. A computer can compare signals mathematically, but that does not always reflect what our ears actually perceive. As a result, listening and comparing by ear remained an important part of the evaluation, a reminder that, at least for now, the human touch still matters.

To complement the project, a simple computer program was also created so that curious listeners can test it for themselves and hear how their favourite song would sound inside the Santi Marcellino and Pietro church in Cremona.